THE INDIAN INTERNATIONAL SCHOOL,DSO

Worksheet – Chapter 11: Three Dimensional Geometry (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	The direction cosines of a line equally inclined to the coordinate axes are: (A) $\pm 1, \pm 1, \pm 1$ (B) $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$ (C) $\pm\frac{1}{3}, \pm\frac{1}{3}, \pm\frac{1}{3}$ (D) $\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}$	1
2.	If a line makes angles $90°$, $135°$ and $45°$ with the x, y and z axes respectively, its direction cosines are: (A) $0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ (B) $0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ (C) $1, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ (D) $0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$	1
3.	The Cartesian equation of the line through $(5, -2, 4)$ and parallel to $3\hat{\imath} + 2\hat{\jmath} - 8\hat{k}$ is: (A) $\frac{x+5}{3} = \frac{y-2}{2} = \frac{z+4}{-8}$ (B) $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{-8}$ (C) $\frac{x-3}{5} = \frac{y-2}{-2} = \frac{z+8}{4}$ (D) $\frac{x-5}{3} = \frac{y+2}{2} = \frac{z-4}{8}$	1
4.	The angle between the lines $\frac{x-2}{2} = \frac{y-1}{5} = \frac{z+3}{-3}$ and $\frac{x+2}{-1} = \frac{y-4}{8} = \frac{z-5}{4}$ is: (A) $\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$ (B) $\cos^{-1}\left(\frac{26}{9}\right)$ (C) $\cos^{-1}\left(\frac{13}{9\sqrt{38}}\right)$ (D) $\frac{\pi}{2}$	1

Q No.	Question	Marks
5.	The value of p for which the lines $\frac{1-x}{3}=\frac{7y-14}{2p}=\frac{z-3}{2}$ and $\frac{7-7x}{3p}=\frac{y-5}{1}=\frac{6-z}{5}$ are perpendicular is: (A) $\frac{70}{11}$ (B) $\frac{11}{70}$ (C) $-\frac{70}{11}$ (D) 7	1
6.	**Assertion (A):** The line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ passes through the origin. **Reason (R):** Any line of the form $\frac{x}{a}=\frac{y}{b}=\frac{z}{c}$ passes through the origin, since $(0,0,0)$ satisfies it. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Find the vector and Cartesian equations of the line passing through $A(1,2,-1)$ and $B(2,1,1)$.	2
	LEVEL 2 **Q 8 – Q 14**	**16**
8.	Show that the lines $\frac{x-5}{7}=\frac{y+2}{-5}=\frac{z}{1}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ are perpendicular to each other.	2
9.	Find the shortest distance between the lines $\vec{r}=(\hat{\imath}+2\hat{\jmath}+\hat{k})+\lambda(\hat{\imath}-\hat{\jmath}+\hat{k})$ and $\vec{r}=(2\hat{\imath}-\hat{\jmath}-\hat{k})+\mu(2\hat{\imath}+\hat{\jmath}+2\hat{k})$.	3
10.	Find the distance between the parallel lines $\vec{r}=\hat{\imath}+2\hat{\jmath}-4\hat{k}+\lambda(2\hat{\imath}+3\hat{\jmath}+6\hat{k})$ and $\vec{r}=3\hat{\imath}+3\hat{\jmath}-5\hat{k}+\mu(2\hat{\imath}+3\hat{\jmath}+6\hat{k})$.	3
11.	Find the angle between the lines $\vec{r}=3\hat{\imath}+2\hat{\jmath}-4\hat{k}+\lambda(\hat{\imath}+2\hat{\jmath}+2\hat{k})$ and $\vec{r}=5\hat{\imath}-2\hat{\jmath}+\mu(3\hat{\imath}+2\hat{\jmath}+6\hat{k})$.	2
12.	Find k so that the lines $\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y-1}{1}=\frac{z-6}{-5}$ are perpendicular.	2
13.	Find the direction cosines of the line whose Cartesian equations are $5x-3=15y+7=3-10z$.	2
14.	Find the Cartesian equation of the line through $(-2,4,-5)$ and parallel to $\frac{x+3}{3}=\frac{y-4}{5}=\frac{z+8}{6}$.	2
	LEVEL 3 **Q 15 – Q 20**	**27**
15.	Find the coordinates of the foot of the perpendicular drawn from $P(1,2,3)$ to the line $\frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}$. Also find the length of the perpendicular.	5
16.	Find the image A' of the point $A(1,6,3)$ in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$. Also, find the equation of the line joining A and A'.	5
17.	Show that the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{3}=\frac{z-6}{5}$ intersect. Find their point of intersection.	5

Q No.	Question	Marks
18.	Find a point P on the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$ such that its distance from the point $Q(2,4,-1)$ is 7 units. Also, find the equation of the line joining P and Q.	5
19.	Find the vector equation of the line passing through $(1,2,-4)$ and perpendicular to both the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.	3
20.	**Case Study:** Two aircraft fly along straight paths given by $L_1: \vec{r} = (\hat{\imath} + \hat{\jmath}) + \lambda(2\hat{\imath} - \hat{\jmath} + \hat{k})$ and $L_2: \vec{r} = (2\hat{\imath} + \hat{\jmath} - \hat{k}) + \mu(3\hat{\imath} - 5\hat{\jmath} + 2\hat{k})$. Air-traffic control needs to know how close the paths come. **(i)** Write the direction ratios of L_1 and L_2. **[1]** **(ii)** Are the paths parallel? Justify. **[1]** **(iii)** Find the shortest distance between the two paths. **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 11: Three Dimensional Geometry

Q No.	Answer / Key Steps
	LEVEL 1
1.	(B)
2.	(B) $\cos 90°, \cos 135°, \cos 45°$
3.	(B)
4.	(A) $\cos\theta = \frac{\lvert -2+40-12 \rvert}{\sqrt{38}\cdot 9} = \frac{26}{9\sqrt{38}}$
5.	(A) DRs $\left(-3, \frac{2p}{7}, 2\right)$ and $\left(-\frac{3p}{7}, 1, -5\right)$; $\frac{11p}{7} - 10 = 0$
6.	(A) Both true; R explains A.
7.	$\vec{r} = \left(\hat{\imath} + 2\hat{\jmath} - \hat{k}\right) + \lambda\left(\hat{\imath} - \hat{\jmath} + 2\hat{k}\right)$; $\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z+1}{2}$
	LEVEL 2
8.	$7(1) + (-5)(2) + 1(3) = 0$
9.	$\vec{b}_1 \times \vec{b}_2 = -3\hat{\imath} + 3\hat{k}$, $(\vec{a}_2 - \vec{a}_1) = \hat{\imath} - 3\hat{\jmath} - 2\hat{k}$; SD $= \frac{\lvert -3-6 \rvert}{3\sqrt{2}} = \frac{3}{\sqrt{2}}$ units
10.	$\vec{b} \times (\vec{a}_2 - \vec{a}_1) = -9\hat{\imath} + 14\hat{\jmath} - 4\hat{k}$; $d = \frac{\sqrt{293}}{7}$ units
11.	$\cos\theta = \frac{3+4+12}{3\times 7} = \frac{19}{21}$; $\theta = \cos^{-1}\frac{19}{21}$
12.	$-9k + 2k - 10 = 0 \Rightarrow k = -\frac{10}{7}$
13.	[PT2 Q24] $\frac{x-\frac{3}{5}}{\frac{1}{5}} = \frac{y+\frac{7}{15}}{\frac{1}{15}} = \frac{z-\frac{3}{10}}{-\frac{1}{10}}$; DRs $6,2,-3$; DCs $\frac{6}{7}, \frac{2}{7}, -\frac{3}{7}$
14.	$\frac{x+2}{3} = \frac{y-4}{5} = \frac{z+5}{6}$
	LEVEL 3
15.	General point $(3\lambda + 6, 2\lambda + 7, -2\lambda + 7)$; $\lambda = -1$. Foot $(3,5,9)$; length $= 7$ units
16.	Foot $(1,3,5)$; image $A'(1,0,7)$. Line AA': $\frac{x-1}{0} = \frac{y-6}{-3} = \frac{z-3}{2}$
17.	Equating general points: $\lambda = \frac{1}{2}$, $\mu = -\frac{3}{2}$ satisfy all three; point $\left(\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right)$
18.	$P(\lambda - 5, 4\lambda - 3, 6 - 9\lambda)$; $PQ^2 = 49 \Rightarrow 98(\lambda - 1)^2 = 0 \Rightarrow \lambda = 1$; $P(-4,1,-3)$. Line PQ: $\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$
19.	$\vec{b}_1 \times \vec{b}_2 = 24\hat{\imath} + 36\hat{\jmath} + 72\hat{k} \parallel 2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}$; $\vec{r} = \left(\hat{\imath} + 2\hat{\jmath} - 4\hat{k}\right) + \lambda\left(2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}\right)$
20.	(i) $2, -1, 1$ and $3, -5, 2$. (ii) No; DRs are not proportional. (iii) $\vec{b}_1 \times \vec{b}_2 = 3\hat{\imath} - \hat{\jmath} - 7\hat{k}$, $\vec{a}_2 - \vec{a}_1 = \hat{\imath} - \hat{k}$; SD $= \frac{10}{\sqrt{59}}$ units